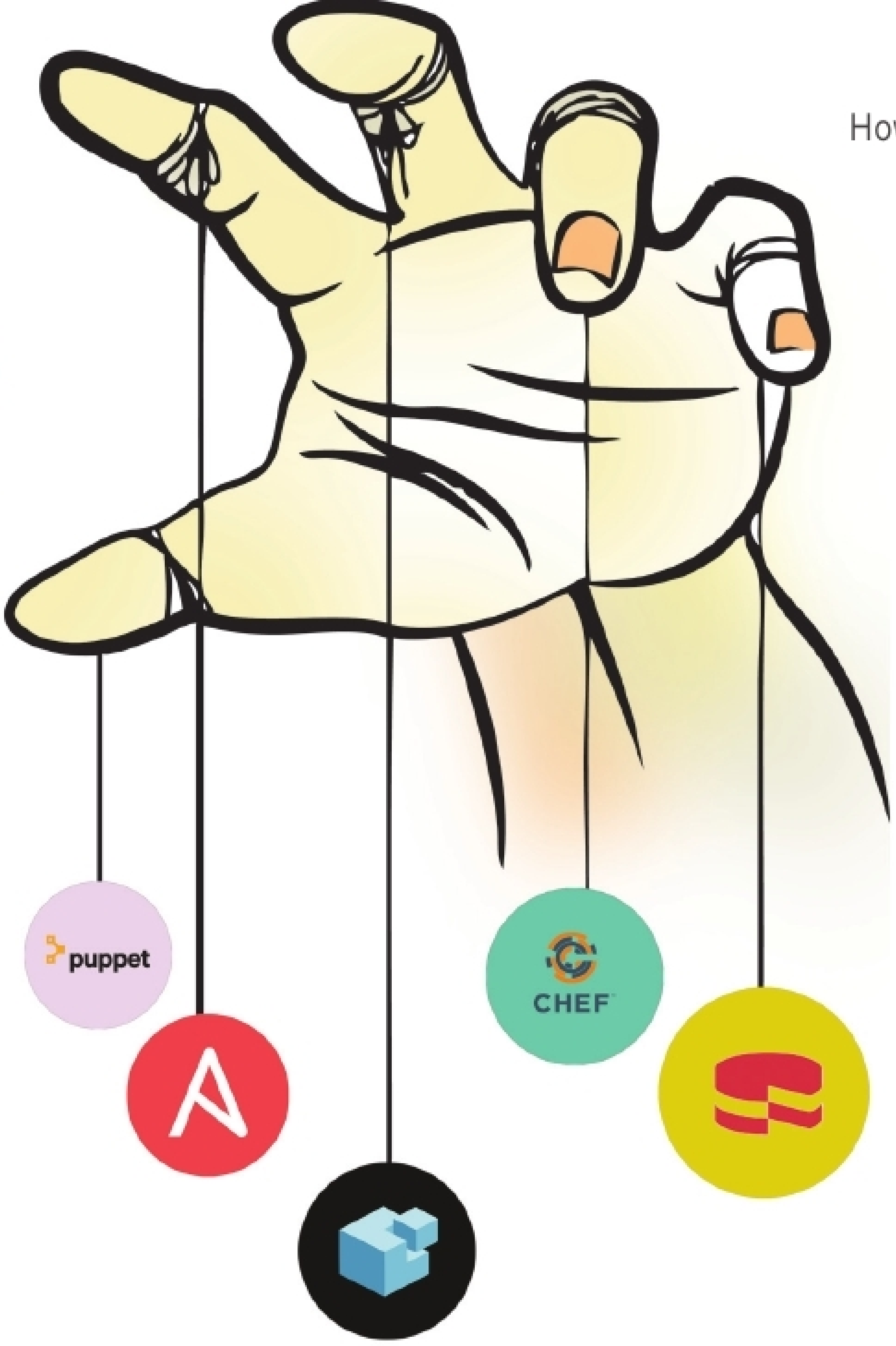
Automating Infrastructure Management with Puppet



Maintaining infrastructure configuration in the optimal state is a complex task. The heterogeneity in software and hardware configurations makes the task difficult. Puppet enables the seamless automation of infrastructure management, thus reducing the burden on the systems administrator . This article introduces you to the features of Puppet.

With the prolific increase in the heterogeneity of components in IT infrastructure, managing this environment efficiently becomes an important task. Systems administrators go through this complex routine of configuring and deploying servers. These daily tasks, though repetitive, require expert skills owing to the changes and enhancements in the infrastructure. Moreover, the standalone scripts that are custom made to automate these tasks suffer from various disadvantages like a lack of proper documentation, improper updates, platform specific requirements, etc.

One of the crucial problems that many IT organisations face is the differences in the development, testing and production environments. The code that works perfectly in the development environment may induce some glitches in the QA or production environment. This becomes a big barrier when introducing new features in the production environment. Even the bug fixes become cumbersome. Ultimately, it takes longer to make the best features reach the end users. This becomes a big issue in the present scenario of cut-throat competition.

There are certain tools that can make the complex task of infrastructure configuration management effective and efficient, which are listed below:

- Puppet
- SaltStack
- Ansible
- Chef
- Cake

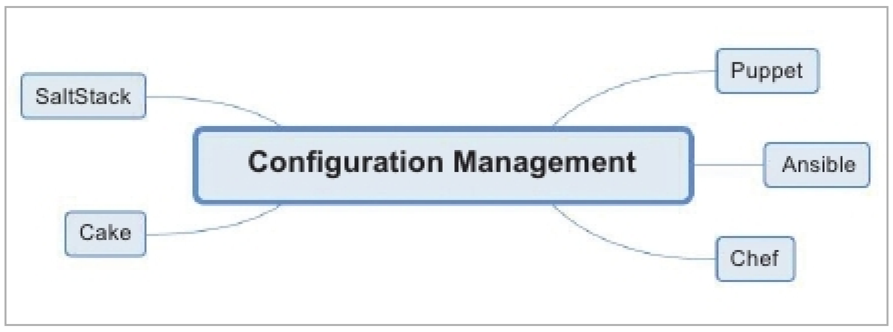


Figure 1: Configuration management